

Fleet Analytics & Risk Prioritization

Souradeep Roy

souro26.ju@gmail.com

1. Executive Summary

This project analyzes fleet telemetry to support operational decision-making.

It focuses on two questions: which drivers show relatively unusual behaviour and should be reviewed, and which vehicles show unusual sensor behaviour that should be considered for inspection.

My main decision was to measure risk relative to the fleet instead of creating outside thresholds that the data could not justify. I converted the selected telemetry metrics to scores from 0 to 10. A higher score therefore means that the observation is more extreme compared with the other drivers or vehicles in this dataset.

For drivers, I used speed, acceleration and manoeuvring behaviour. For vehicles, I used horizontal acceleration and gyroscope behaviour. I gave the final components equal weight because I did not have a defensible reason to say that one component should count more than another.

I also tested whether the rankings changed too much when I changed the weights or removed individual metrics. This was important because I did not want a ranking that looked convincing only because of one arbitrary modelling choice.

The final outputs are two Power BI dashboards. They let a stakeholder first identify a driver or vehicle that needs attention and then look at the individual components behind its score. I treat the results as a screening and prioritization system. A high score is a reason to investigate something further. It is not proof that a driver is unsafe or that a vehicle has a mechanical fault.

2. Business Problem and Objectives

I approached the project as a decision problem rather than just a data description. The useful output is not a list of telemetry statistics. I wanted the analysis to help answer who or what should receive attention first and give enough information to understand why.

For driver behaviour, I wanted to identify drivers whose observed behaviour was relatively extreme within the fleet and identify the parts of their behaviour that contributed to the score. For vehicle health, I wanted to identify vehicles with unusual sensor signatures and provide a reasonable basis for inspection priority.

I therefore aimed to build interpretable driver and vehicle scores, check their robustness, and present them through dashboards that a stakeholder could use without having to read the underlying notebooks.

3. Dataset and Data Preparation

I worked with four related tables: Drivers, Vehicles, Trips and Telemetry. Drivers and Vehicles provide the entities being analysed. Trips connect those entities to individual journeys, and Telemetry provides the sensor observations recorded during those journeys.

The dataset contains 30 drivers, 30 vehicles and 450 trips. I used the trip structure to connect telemetry to both the driver and vehicle involved in each trip. This matters because I did not want to treat all telemetry as one undifferentiated pool.

I first prepared and checked the data before building the scoring models. I checked the table structures, identifiers, relationships, missing values and duplicates. I also derived the sensor quantities needed for the analysis and prepared processed data for the driver and vehicle notebooks.

I kept driver and vehicle analysis separate. A vehicle showing an unusual sensor pattern should not automatically be blamed on its driver, and a driver's behavioural score should not be treated as evidence of a mechanical problem with the vehicle.

4. Driver Behaviour Methodology

I selected five metrics because I wanted the score to cover different aspects of driving behaviour rather than relying on one variable.

Component	Metrics	Why I used them
Speed Risk	P95 Speed Extreme Speed Rate	I wanted both the upper range of speed and the frequency of unusually high speed.
Acceleration Risk	P95 Horizontal Acceleration	I wanted to capture unusually strong acceleration while reducing the influence of isolated outliers.
Manoeuvring Risk	P95 Gyroscope Magnitude Extreme Gyroscope Rate	I wanted both the magnitude and frequency of unusual rotational behaviour.

I converted each metric to a fleet relative score from 0 to 10. I chose this instead of a fixed safety threshold because the dataset did not give me an externally validated threshold. The score therefore answers how extreme a driver's behaviour is relative to this fleet, not whether the driver has crossed an objective safety limit.

I then combined the three component scores using equal weighting:

$$\text{Driver Risk Score} = (\text{Speed Risk} + \text{Acceleration Risk} + \text{Manoeuvring Risk}) / 3$$

I used equal weighting because I did not have enough evidence to justify giving speed, acceleration or manoeuvring a different importance. It also keeps the score easy to explain. If a driver has a high final score, I can still show which of the three components is responsible.

I tested alternative weights and removed individual metrics to see whether the rankings were overly dependent on one assumption. I also used driver level leave one out checks. The purpose was not to claim that one scoring formula is objectively correct. The purpose was to make sure the main conclusions did not disappear under reasonable changes.

5. Driver Behaviour Results

V23 had the highest Driver Risk Score at 9.66. V06, V14, V03 and V19 followed at approximately 8.51. V09, V11 and V22 were at the lower end, each at approximately 0.46.

The important point for me was that V23 remained the highest ranked driver when I changed the relative weighting of the components. The ordering of some other high risk drivers changed, but the main high and low groups remained broadly identifiable.

The sensitivity checks also showed why I should not interpret the final score by itself. Individual scores can change when a metric is removed, but the overall structure of the ranking remains useful. This is why the dashboard shows the component profile alongside the overall ranking.

I designed the Driver Behaviour dashboard around three questions. Who needs attention? What is contributing to that driver's score? How is risk distributed across the fleet? The ranking answers the first question, the selected driver profile answers the second, and the fleet distribution gives the wider context.

6. Vehicle Health Methodology

For vehicles, I wanted to identify unusual sensor behaviour that could justify a closer inspection. I did not want to call an unusual sensor pattern a mechanical failure because the dataset does not contain confirmed failure labels.

Metric	Why I used it
P95 Horizontal Acceleration	It captures the upper range of unusually strong horizontal acceleration.
P95 Gyroscope Magnitude	It captures the upper range of rotational sensor activity.
High Gyroscope Rate	It captures how frequently high rotational activity occurs, complementing the P95 measure.

I converted these metrics to 0 to 10 fleet relative scores and combined them with equal weighting:

$$\text{Inspection Score} = (\text{Acceleration Score} + \text{Gyroscope Score} + \text{High Gyroscope Rate Score}) / 3$$

Again, the equal weighting is a modelling assumption. I used it because the available data did not give me a sound reason to make one sensor dimension dominate the others. The result is therefore an inspection priority score, not a diagnosis.

I tested alternative weights and removed each sensor component individually. I also looked at driver assignment because I did not want to assume that a vehicle anomaly was caused by one driver. V23, the highest ranked vehicle, was associated with multiple drivers. That makes a single driver explanation less convincing, although it still does not prove that the vehicle has a mechanical problem.

7. Vehicle Health Results

V23 had the highest Inspection Score at 9.66. V14, V19, V06 and V03 followed at approximately 8.51. The leading group remained broadly identifiable when I changed the weighting or removed individual sensor metrics.

I therefore use the ranking as a practical way to decide which vehicles should be looked at first. I do not use it as a statement that those vehicles have failed or are definitely unsafe. The component scores provide the next level of information by showing whether acceleration, gyroscope activity or high gyroscope activity is driving the priority.

The Vehicle Health dashboard follows the same logic. I first show which vehicles deserve attention. I then show the selected vehicle's anomaly components. I also show the fleet distribution and the relationship between sensor measures so that the stakeholder can judge the selected vehicle against the rest of the fleet.

8. Key Findings and Business Recommendations

The strongest practical finding is that the telemetry can be reduced to a relatively small set of interpretable priorities without hiding the information behind the composite score. V23 stands out in both the driver and vehicle analysis, while the other high ranked items form a smaller group that can be reviewed in order of priority.

For drivers, I would use the ranking to prioritize behavioural review rather than treating the ranking as a final judgement. The component scores can then guide the review toward speed, acceleration or manoeuvring behaviour.

For vehicles, I would use the Inspection Score to decide where maintenance attention is most worth spending first. A high score should lead to a closer inspection or review of the relevant telemetry, especially when more than one sensor indicator is elevated.

The dashboards are therefore designed to narrow the search space. They do not replace a maintenance inspection or a human review. They help decide where that review should start.

9. Limitations

I treat the scores as relative indicators because the dataset does not provide validated external thresholds. A score of 9 therefore means that the observation is near the extreme end of this fleet, not that it has crossed a universal safety limit.

The vehicle analysis does not contain confirmed mechanical failure labels. A high Inspection Score is consequently a reason to inspect a vehicle, not proof of a fault.

The available observation period also limits what I can say about persistence. I can identify patterns in the supplied data, but I cannot claim that a pattern will continue over a much longer period.

Accelerometer and gyroscope measurements can be affected by sensor placement, mounting, measurement noise and operating conditions. These factors can contribute to an unusual reading without implying a mechanical problem.

Driver and vehicle assignments are not one to one. I therefore cannot automatically attribute a vehicle anomaly to a driver. The V23 driver assignment check is supporting evidence, not a causal test.

Finally, the scores depend on the features and equal weighting assumptions I selected. I addressed this by running sensitivity and metric removal checks, but those checks do not make the scoring system objectively correct. They only show how stable the main conclusions are under the alternatives I tested.

I would use the results as a first level filter. If a driver or vehicle ranks highly, I would investigate the underlying behaviour or sensor readings before making a final decision. The score helps me decide where to look first, but it does not make that final decision for me.